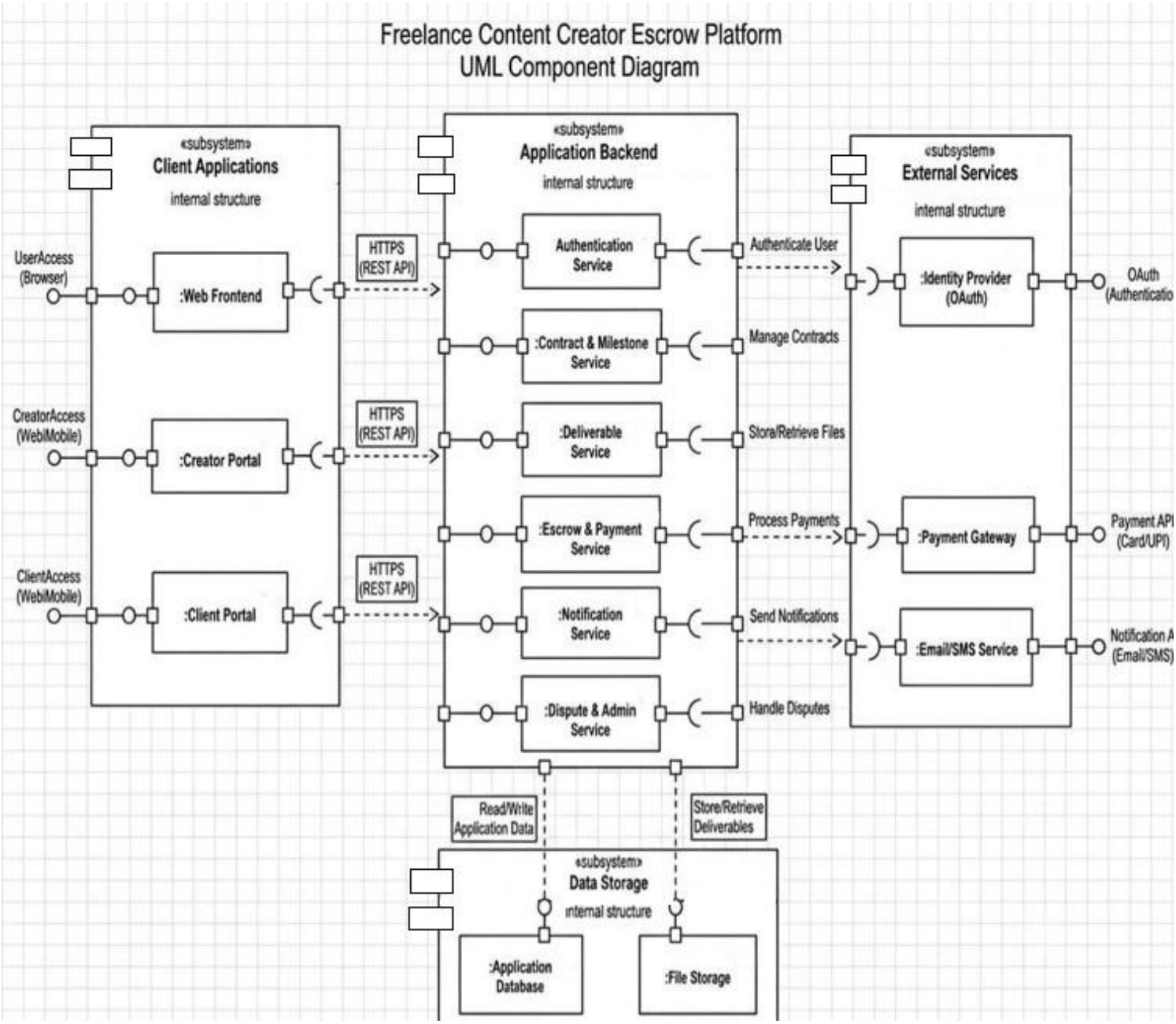
Lab 3: Component Modelling & Architectural Pattern Selection

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Freelance Content Creator Escrow Platform

1.COMPONENT DIAGRAM



2. ARCHITECTURAL JUSTIFICATION

Architecture Selection: We chose the Microservices Architecture for the Freelance Content Creator Escrow Platform.

Architectural Choice: The system is decomposed into independently deployable backend services — Authentication, Contract & Milestone, Deliverable, Escrow & Payment, Notification, and Dispute & Admin — each exposing a well-defined REST interface. Client applications (Web Frontend, Creator Portal, Client Portal) consume these services over HTTPS, while third-party providers (OAuth identity provider, payment gateway, email/SMS gateway) are reached through required interfaces. Persistence is separated into an application database and a file store for deliverables.

Two Reasons for the Choice

- **Independent scaling of uneven workloads.** The workload on this platform is highly asymmetric. Deliverable upload/download and notification traffic are heavy and bursty, whereas contract creation and dispute handling are comparatively rare. Microservices let the Deliverable and Notification services be scaled out on their own without provisioning extra capacity for the entire backend, which a layered or single-server client-server design would force us to do.
- **Fault isolation around money and trust.** Escrow is the core promise of the platform: funds must be held safely and released only on milestone approval. Isolating the Escrow & Payment service means a failure in an unrelated area — a slow file upload, an SMS provider outage, a dispute-admin bug — cannot crash or corrupt the payment path. Teams can also release changes to contracts, disputes or notifications without redeploying the payment service.

Security Advantage

Security is enforced at service boundaries rather than inside one large process. All client traffic arrives over HTTPS and must first obtain a token from the Authentication service, which delegates identity verification to an external OAuth provider; every other service independently validates that token before acting. Because sensitive concerns are separated, card data never enters our own components — the Escrow & Payment service delegates to the payment gateway over a dedicated required interface — which shrinks the attack surface and keeps payment-card scope confined to a single service. A compromise of a low-privilege component such as Notification therefore does not expose contract, escrow or identity data.

Performance Benefit

Services communicate asynchronously where latency matters, so slow operations do not block user-facing ones. A creator uploading a large video interacts only with the Deliverable service and file storage, while the client's browser continues to browse contracts through a separate service and separate database connections. Notifications are dispatched out-of-band instead of holding the request open until the email or SMS provider responds. Combined with per-service horizontal scaling, this keeps response times for milestone approval and payment release low even during peak upload periods.